Resolving the memory background by energy Conserved subtraction and the first fluctuation pair

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Abstract

The scalar mean of the temporal-singlet memory is insufficient as a fluctuation counterterm. We determine its limiting regression against energy functions using the established q-Mehler Markov operator. An explicit positive variance gives a lower bound showing why scalar centering fails at the collective normalization. Exact conserved polynomials instead yield a finite pair of memory coefficients. Their complete two-energy spectral matrix follows from the previously computed quadratic, cubic and rank-filtered singlet channels. Energy centering removes the time descendant present in the raw cubic insertion; the normalized memory tangent tends to the cubic sech^6 kernel at small coupling.

1 What the energy-weighted mean determines

Use the same standard CAR complex p-body Gaussian Hamiltonian $H = H_s$, including overlapping index sets, with $w = 4^p(J_s^2 N/p^2)(p!/N^p)^2$. Brackets denote the annealed full normalized trace at $\beta = \mu = 0$, and $\hbar = 1$. First take $N, p \to \infty$, $p^2/N \to \lambda > 0$, then optionally $\lambda \downarrow 0$. Set $q = e^{-4\lambda}$, $\sigma^2 = J_s^2/\lambda$; all times use string units.

The preceding stage defined, in each nonzero charge block,

$$h = H_m, \quad K = 1_N \otimes H_{m-1}, \quad V\psi = m^{-1/2} \sum_i |i\rangle c_i \psi, \quad Q = 1 - VV^\dagger, \tag{1}$$

and $\mathcal{R} = QKV$, $K_\perp = QKQ$. The energy-centered memory is

$$\widehat{\mathcal{M}}_m(s) = e^{-ihs/2} \mathcal{R}^\dagger e^{iK_\perp s} \mathcal{R} e^{-ihs/2}, \qquad \widehat{\mathcal{M}}_0(s) = 0. \tag{2}$$

Its scalar mean satisfies $p\langle \widehat{\mathcal{M}}(s) \rangle \to 4J_s^2 G_q(s)$. For every bounded continuous energy test f, the stronger statement is

$$p\langle f(H)^\dagger \widehat{\mathcal{M}}(s) \rangle \longrightarrow 4J_s^2 \int \overline{f(x)} g_s(x)\, d\nu_q(x), \qquad g_s(x) = e^{-isx}(\mathsf{P}_q e^{is\cdot})(x). \tag{3}$$

Here ν_q is the variance-σ^2 Hamiltonian law and P_r is the established Mehler transition operator.

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2 A nonconstant conserved background

Write $d\mu_r(x,y) = d\nu_q(x)\,\mathsf{P}_r(x,dy)$, using the same normalized two-energy probe measure as in the preceding stage. For monic q-Hermite polynomials with $H_{n+1}(x) = xH_n(x) - \sigma^2[n]_q H_{n-1}(x)$, the known Markov operator obeys [1]

$$\mathsf{P}_r H_n = r^n H_n, \qquad \mathsf{P}_r^* = \mathsf{P}_r, \qquad \mathsf{P}_r\mathsf{P}_t = \mathsf{P}_{rt}. \tag{4}$$

These are energy-space transition operators. Their parameter is not the real memory time s. The Hamiltonian support is $[-2\sigma/\sqrt{1-q}, 2\sigma/\sqrt{1-q}]$.

To prove (3), insert $f(h)^\dagger$ into the two memory–force comparison bounds from the preceding note. Their right-hand sides acquire at most $\|f\|_\infty$. The charge-window and tail estimates are unchanged. The resulting force expression is an integral of $\overline{f(x)}e^{is(y-x)}$ against its positive two-energy measure, which converges to $4J_s^2\mu_q$. This proves the stated weighted mean. It does not prove strong operator convergence to $4J_s^2 g_s(H)$.

Nevertheless the nonconstant background is measurable exactly:

$$\int g_s\, d\nu_q = G_q(s), \tag{5}$$

$$V_{\mathrm{bg}}(s) := \int |g_s - G_q(s)|^2\, d\nu_q = G_{q^2}(s) - G_q(s)^2 \geq 0. \tag{6}$$

Indeed multiplication by e^{-isx} preserves the norm, and $\|\mathsf{P}_q e^{isx}\|_2^2 = \langle e^{isx}, \mathsf{P}_{q^2} e^{isx}\rangle_{L^2(\nu_q)} = G_{q^2}(s)$. The kernels are real by simultaneous energy reflection. Low moments give

$$g_s(x) = 1 + is(q-1)x - \tfrac{1}{2}s^2[(1-q)^2x^2 + (1-q^2)\sigma^2] + O(s^3), \tag{7}$$

$$V_{\mathrm{bg}}(s) = \sigma^2(1-q)^2 s^2 + O(s^4). \tag{8}$$

Thus it is strictly positive for sufficiently small nonzero s.

Let $X_N = p\widehat{\mathcal{M}}(s)$ and $\|B\|_2^2 = \langle B^\dagger B\rangle$. Taking the test $g_s - G_q(s)$ in (3), with a bounded continuous extension outside the limiting support, and applying Cauchy–Schwarz,

$$\liminf_{\mathrm{DS}} \|X_N - \langle X_N\rangle 1\|_2^2 \geq 16 J_s^4 V_{\mathrm{bg}}(s). \tag{9}$$

The full-trace charge number is Q_f. Central-window concentration and $\|\widehat{\mathcal{M}}\|_2^2 \leq \langle G^2\rangle = O(p^{-2})$ give

$$\liminf_{\mathrm{DS}} \frac{\|Q_f\widehat{\mathcal{M}} - \langle Q_f\widehat{\mathcal{M}}\rangle 1\|_2^2}{p^2} \geq \frac{4J_s^4}{\lambda^2} V_{\mathrm{bg}}(s). \tag{10}$$

Scalar centering therefore leaves a divergent collective background.

3 Exact subtraction at the first two memory orders

At finite N, set $r = p/m$ and expand

$$\widehat{\mathcal{M}}_m(s) = G + isZ_1 + O(s^2), \qquad G = A_m/m - r^2h^2, \qquad mZ_1 = T_{3,m} + 2r\mathcal{J}_hA_m - mr^3h^3. \tag{11}$$

Here $A = \sum_i [H, c_i]^\dagger [H, c_i]$, $M_3 = \sum_i c_i^\dagger \operatorname{ad}_H^3 c_i$, $T_3 = (M_3 + M_3^\dagger)/2$, and $\mathcal{J}_hB = (hB + Bh)/2$. Let $C_m = \langle h^2 \rangle_m$, $a_m = \langle A_m \rangle_m$, and use the previously defined ensemble projection coefficient

$$c_m = \frac{\langle (A_m - a_m)(h^2 - C_m) \rangle_m}{\langle (h^2 - C_m)^2 \rangle_m}, \qquad Y_m = A_m - a_m - c_m(h^2 - C_m). \tag{12}$$

All scalar coefficients here are deterministic charge-block functions. For the already calculated cubic channels put

$$a = -\frac{(1-q)^2}{1+q^2}, \qquad b = \frac{2\lambda(1-q)(1+q^3)}{(1+q^2)(1+q+q^2)}, \tag{13}$$

$$R = W_3T_3 - aW_3(\mathcal{J}_hA) - bW_3h^3, \tag{14}$$

$$F = W_1T_3 - \bar{t}_m h, \qquad B_S = W_1(\mathcal{J}_hA) - \bar{s}_m h, \quad B_Z = W_1h^3 - \bar{z}_m h. \tag{15}$$

W_j denotes ordinary Gaussian coupling chaos; bars are projections of the respective linear-chaos operator onto h. The earlier exact cubic reduction gives

$$P_m = (ac_m + b)h^3 + [\bar{t}_m - (ac_m + b)\bar{z}_m]h, \tag{16}$$

$$T_{3,m} - P_m = R + a\mathcal{J}_hY_m + F - aB_S - bB_Z. \tag{17}$$

The required two conserved counterterms are consequently explicit:

$$C_{0,m} = a_m + c_m(h^2 - C_m) - mr^2h^2, \tag{18}$$

$$C_{1,m} = P_m + 2r[a_mh + c_m(h^3 - C_mh)] - mr^3h^3. \tag{19}$$

Both commute with h. Subtracting them gives the exact pair

$$X_{0,N} = mG - C_{0,m} = Y_m, \tag{20}$$

$$X_{1,N} = mZ_1 - C_{1,m} = R + (a + 2r)\mathcal{J}_hY_m + F - aB_S - bB_Z. \tag{21}$$

At fixed λ, the previous rank-tail bounds imply $B_S, B_Z = O_{L^2}(p^{-1})$ and $r\mathcal{J}_hY \to 0$. Hence the second coefficient has limiting representative $L = R + a\mathcal{J}_HY + F$. This names a limit of moments and two-point spectra, not an identification of Hilbert spaces at different N.

4 The full two-energy spectral matrix of the pair

Use the established vanishing cross measures for F, Y, R and their diagonal measures $V_F\mu_q$, $W\mu_{q^2}$, $V_R\mu_{q^3}$. For $e = (x+y)/2$ the pair (Y, L) has the positive spectral matrix

$$d\boldsymbol{\nu} = \begin{pmatrix} W\,d\mu_{q^2} & aeW\,d\mu_{q^2} \\ aeW\,d\mu_{q^2} & V_F\,d\mu_q + V_R\,d\mu_{q^3} + a^2e^2W\,d\mu_{q^2} \end{pmatrix}. \tag{22}$$

Relative to the product Hamiltonian measure, its determinant equals $WK_{q^2}(V_FK_q + V_RK_{q^3}) \geq 0$. The coefficients are

$$W = 4J_s^4\left(\lambda^{-1} + \frac{4q}{1+q}\right), \qquad V_F = 64J_s^6q^2, \tag{23}$$

$$V_R = \sigma^6\frac{\lambda A_1 + \lambda^2 A_2}{D^2}, \qquad D = (1+q^2)(1+q+q^2), \tag{24}$$

$$A_1 = 4(1-q^2)(1+q^2)(1+q+q^2)^2,$$

$$A_2 = 16q^2(1+q)(1+q^2)(1+q+q^2)(2q^2-q+2).$$

These diagonal channel results are inherited from the preceding singlet calculations, not recalculated as new probe formulas. The application here is the exact memory-coefficient combination.

At any external centre time T, the off-diagonal ordinary correlator vanishes. The transformation $(x, y) \mapsto (-y, -x)$ preserves $y - x$ and the measure but reverses e. Therefore

$$\langle Y^\dagger L(T)\rangle_{\mathrm{DS}} = 0, \tag{25}$$

$$\langle Y^\dagger Y(T)\rangle_{\mathrm{DS}} = WG_{q^2}(T), \tag{26}$$

$$\langle L^\dagger L(T)\rangle_{\mathrm{DS}} = V_FG_q(T) + V_RG_{q^3}(T) + a^2WJ_2(T), \tag{27}$$

$$J_2(T) = \int e^2e^{iT(y-x)}\,d\mu_{q^2}(x, y). \tag{28}$$

Ordinary-time orthogonality does not erase the nonzero energy-resolved cross entry in (22). Its energy-weighted integral is $aW\sigma^2(1+q^2)/2$.

The first-degree memory polynomial $\Xi(s) = Y + isL$ now has the complete centre-time covariance

$$\langle \Xi(s)^\dagger\Xi(u; T)\rangle = WG_{q^2}(T) + su\,[V_FG_q(T) + V_RG_{q^3}(T) + a^2WJ_2(T)]. \tag{29}$$

This is an exact covariance of the limiting pair. For the actual finite-memory field it specifies the $(0,0), (1,0), (0,1), (1,1)$ Taylor coefficients, with derivatives taken before double scaling. Terms of degree two in either memory argument remain uncomputed.

5 What energy centering selects at small coupling

The original non-Hermitian derivative has

$$M_3 - P_m \longrightarrow R + a\mathcal{J}_H Y + F + \tfrac{3}{2}[H, Y]. \tag{30}$$

The memory tangent L has no final time-descendant term. This follows from the exact centered memory expansion, not from manually discarding a large contribution to the raw cubic spectrum.

Its squared norm and second frequency moment are

$$N_L = V_R + V_F + a^2 W\sigma^2(1+q^2)/2, \tag{31}$$

$$D_L = 2\sigma^2\{V_R(1-q^3) + V_F(1-q)\} + \tfrac{1}{2}a^2 W\sigma^4(1+q)(1-q^4). \tag{32}$$

In the subsequent $\lambda \downarrow 0$ limit,

$$\lambda W \to 4J_s^4, \qquad \lambda N_L \to 32J_s^6, \qquad D_L/N_L \to 24J_s^2. \tag{33}$$

The F fraction is $2\lambda + o(\lambda)$; the energy-descendant fraction is $8\lambda^3 + o(\lambda^3)$. Thus

$$\frac{\langle L^\dagger L(T)\rangle}{N_L} \longrightarrow \operatorname{sech}^6(2J_sT), \qquad \frac{\langle Y^\dagger Y(T)\rangle}{W} \longrightarrow \operatorname{sech}^4(2J_sT). \tag{34}$$

This applies the established small-coupling probe kernels to the specific memory coefficients. The two labels do not by themselves determine bulk meson masses.

6 Verification and actual finite-coupling effects

Three finite CAR systems verify five exact rational-complex matrix identities each: the two subtractions, conservation of both counterterms, and Hermiticity of the tangent. The calculation obtains the first-return coefficient directly from $V^\dagger K^n V$ and obtains the cubic chaos components separately by Gaussian polarization. Twenty-eight exact polynomial tests verify the conditional q-Hermite action through total degree six; symbolic Laurent limits give $4, 32, 768$ for $J_s = 1$ and the three unnormalized limits above.

For four positive couplings we evaluate the background variance and tangent kernel by Jacobi evolution, with a larger truncation as a check. At $\lambda = 0.25, 0.1$, independent 128/192-node Mehler integration also checks the conditional variance directly, the zero ordinary cross and nonzero energy-weighted cross, and the tangent kernel. Finite-λ differences from sech^6 are retained; they are not numerical errors.

One initial independent comparison failed: the quadrature code had omitted K_{q^2} on the e^2 descendant term, giving a nonunit normalized tangent already at $T = 0$. The original source and values are preserved. Restoring that factor fixes the formula; no tolerance was relaxed. Floating-point refinement is not interval certification.

7 From the tangent pair to finite memory

The next unknown is the full covariance after conserved subtraction, with two finite memory parameters and an independent centre time. The present work determines its first pair of coefficients and proves that a purely scalar subtraction cannot yield the desired scaling. Equation (3) also gives the full leading energy-dependent counterterm as a regression functional, at every fixed memory time.

Three requirements now have a concrete form. First, compute the next centered return coefficient, including the G^2 return term and the fourth derivative singlet, then determine its conserved polynomial and its mixed spectra with Y, L. Second, control the operator remainder with two memory insertions at the collective normalization. The single-insertion comparison used for the mean does not provide this bound. Third, match the resulting microscopic kernel with the bulk internal collective operator and spatial insertions, preserving the distinction between memory time, centre time and a light-front quark fraction.

The energy-dependent regression g_s alone is not the full finite-N conditional expectation onto the commutant. Nor does subtracting its leading $1/p$ term automatically fix all subleading conserved pieces at the $m \sim p^2$ normalization. Equations (19)–(21) show explicitly how such pieces are fixed at the first two orders. At those orders the limiting spectra are absolutely continuous and have no zero-frequency atom. Finite-memory tightness remains an additional problem.

The complete fluctuation field and the spatial holographic dictionary remain open. The useful advance is a specified conserved background, an obstruction to scalar centering, and a positive spectral matrix for the actual first two memory coefficients.

Local record. Source, exact checks, numerical parameters, dependency versions, initial failure and input hashes are retained in the project stage directory. The PDF is compiled twice and reviewed page by page; its source/output hashes are bound to the local evidence record. No Git commit, push or release archive is created in this stage.

References

[1] M. Bożejko, B. Kümmerer and R. Speicher, *q-Gaussian processes: non-commutative and classical aspects*, arXiv:funct-an/9604010, Theorems 1.10 and 4.6, especially pp. 20–22.

[2] M. Berkooz, V. Narovlansky and H. Raj, *Complex Sachdev–Ye–Kitaev model in the double scaling limit*, arXiv:2006.13983v2.

[3] A. Day and Chengheng, local project notes: *From the charged force to the mean memory kernel*; *Conserved-subtracted cubic spectrum*; and the associated singlet-kernel, cubic-cross and filtered-spectrum calculations, 6 November 2026. These supply the model-specific precursor identities and measures used here; no priority claim is made.